

Macroeconomics #5

Endogenous Growth Models

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Today's lecture

- 1 Introduction and Motivation
- 2 The AK Model
- 3 Romer (1990) Variety-Expansion Model
- 4 Scale Effect and the Jones (1995) Model
- 5 R&D Underinvestment and Subsidies
- 6 The Schumpeterian Growth Model

Why Endogenous Growth?

Recall from Solow and Ramsey models:

- Long-run per capita growth is driven entirely by **exogenous** technological progress g
- Diminishing returns to capital $\Rightarrow$ growth eventually stops without technology
- The models are silent on **what drives technological progress**

Key questions:

- Can sustained growth arise *endogenously* from the decisions of agents?
- What is the role of R&D, innovation, and knowledge spillovers?
- Can government policy affect *long-run* growth rates?

- ① **The AK Model:** Eliminate diminishing returns $\Rightarrow$ sustained growth without exogenous technology
- ② **Romer (1990) Model:** Endogenous technological change through R&D and variety expansion
- ③ **Scale Effect and Jones (1995):** Critique and semi-endogenous growth
- ④ **R&D Underinvestment and Subsidies:** Market failure and policy

ADVANCED MACRO- ECONOMICS

An Introduction for
Undergraduates

Angus C. Chu

Foreword by Guido Cozzi



AK

The AK Model: Setup

Key idea: Remove diminishing returns to the *accumulable* factor.

Production function:

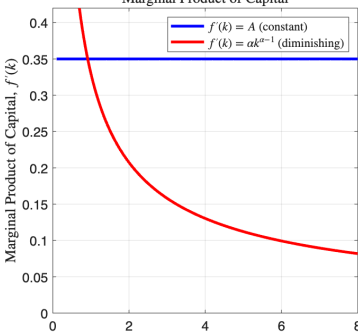
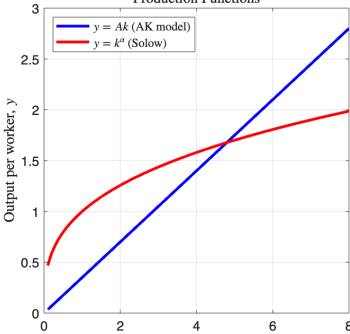
$$Y = AK$$

where $A > 0$ is a constant (technology level) and K is “broad capital” (physical + human).

Key property: The marginal product of capital is **constant**:

$$f'(k) = A \quad \text{for all } k$$

- No diminishing returns $\Rightarrow$ investment remains productive at all levels of K
- Contrast with Solow: $f'(k) = \alpha k^{\alpha-1} \rightarrow 0$ as $k \rightarrow \infty$



Household Problem

Representative household maximizes:

$$U = \int_0^{\infty} e^{-\rho t} \frac{C_t^{1-\theta} - 1}{1-\theta} dt$$

subject to the budget constraint (in per capita terms):

$$\dot{a}(t) = (r(t) - n) a(t) + w(t) - c(t)$$

Euler equation (same as Ramsey):

$$\frac{\dot{c}}{c} = \frac{r - \rho}{\theta}$$

With competitive factor markets and $Y = AK$:

$$r = A - \delta \quad \Longrightarrow \quad \boxed{\frac{\dot{c}}{c} = \frac{A - \delta - \rho}{\theta}}$$

Equilibrium Dynamics

Capital accumulation (per capita):

$$\dot{k}(t) = (A - \delta - n) k(t) - c(t)$$

Consumption path:

$$c(t) = c(0) e^{\frac{A - \delta - \rho}{\theta} t}$$

Transversality condition requires:

$$A > \delta + \rho > (1 - \theta)(A - \delta) + \theta n + \delta$$

which ensures positive growth and convergence of lifetime utility.

Key result: No transition dynamics!

- Growth rates of c , k , and y are *constant* at all times
- $k(t) = k(0) e^{\frac{A - \delta - \rho}{\theta} t}$
- Initial consumption: $c(0) = \left[\frac{(A - \delta)(\theta - 1) + \rho}{\theta} - n \right] k(0)$

AK Model: Properties

Balanced growth rate:

$$g = \frac{A - \delta - \rho}{\theta}$$

Saving rate:

$$s = \frac{\dot{k}(t) + \delta k(t)}{Ak(t)} = \frac{A - \rho - (1 - \theta)\delta}{\theta A} \equiv \text{constant}$$

Contrast with Solow/Ramsey:

- Growth rate depends on **preferences** (ρ, θ) and **technology** (A, δ)
- No convergence: rich countries grow at the same rate as poor countries
- Policy can affect *long-run growth rates*, not just levels

AK Model: Policy Implications

Suppose there is an effective tax rate τ on capital income. The budget constraint becomes:

$$\dot{a}(t) = ((1 - \tau)r(t) - n) a(t) + w(t) - c(t)$$

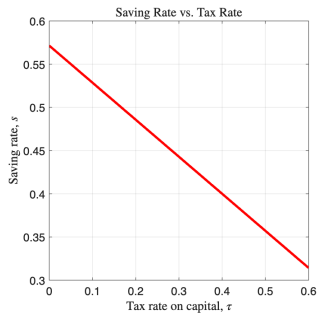
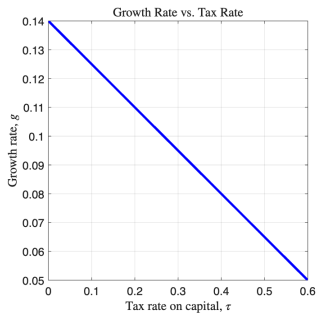
Growth rate of the economy:

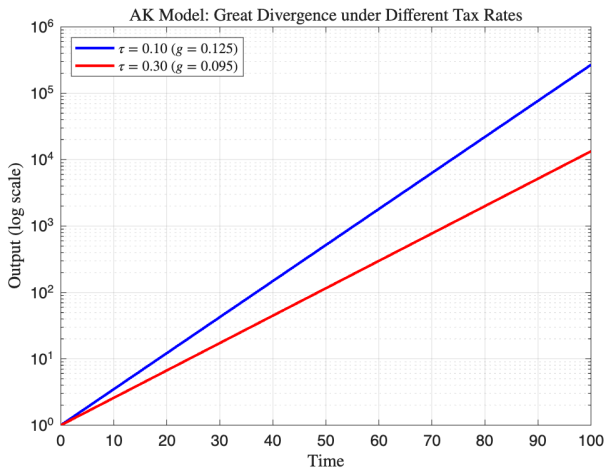
$$g = \frac{(1 - \tau)(A - \delta) - \rho}{\theta}$$

Saving rate:

$$s = \frac{(1 - \tau)A - \rho + \theta n - (1 - \tau - \theta)\delta}{\theta A}$$

- If $A - \delta > 0$, then $\partial s / \partial \tau < 0$: higher taxes $\Rightarrow$ lower savings
- **Unlike Solow**, saving rate responds endogenously to policy
- Different policies lead to **great divergent outcomes**





Strengths:

- Growth is endogenous: determined by parameters and policy
- Simple and tractable

Weaknesses:

- **No transition dynamics**: unrealistic—adjustments happen instantaneously
- **No explicit role for innovation/R&D**: “broad capital” is a black box
- Constant returns to K alone is a knife-edge assumption
- Predicts no convergence at all—too stark

⇒ Need a model that explicitly captures the **innovation process**

Romer Model

Romer (1990): Overview

Paul Romer (1990): “Endogenous Technological Change”

- [Nobel Prize in Economics 2018](#) (shared with Nordhaus)
- Growth is driven by **intentional R&D** by profit-seeking firms

Key mechanism: *Variety expansion*

- Entrepreneurs invest in R&D to invent **new varieties** of intermediate goods
- More varieties $\Rightarrow$ higher productivity $\Rightarrow$ sustained growth

Four sectors:

- ① Household sector
- ② Final good sector (competitive)
- ③ Intermediate goods sector (monopolistic competition)
- ④ R&D sector (free entry)

Household Sector

Representative household with log utility:

$$U = \int_0^{\infty} e^{-\rho t} \ln C_t dt, \quad \rho > 0$$

Subject to the asset accumulation equation:

$$\dot{F}_t = r_t F_t + W_t L - C_t$$

where F_t = financial assets, W_t = wage rate, L = labour supply (inelastic).

Euler equation (log utility):

$$\frac{\dot{C}_t}{C_t} = r_t - \rho$$

On a balanced growth path with constant r :

$$g_C = r - \rho$$

Final Good Sector

A competitive final good sector produces Y_t using labour L_Y and a continuum of intermediate goods $X_t(i)$ for $i \in [0, N_t]$:

$$Y_t = L_Y^{1-\alpha} \int_0^{N_t} X_t(i)^\alpha di, \quad \alpha \in (0, 1)$$

where N_t is the **number of available varieties** at time t .

Profit maximization yields factor demands:

$$W_t = (1 - \alpha) \frac{Y_t}{L_Y} \quad (\text{wage})$$

$$P_t(i) = \alpha L_Y^{1-\alpha} X_t(i)^{\alpha-1} \quad (\text{inverse demand for variety } i)$$

$$\text{so } P_t(i)X_t(i) = \alpha L_Y^{1-\alpha} X_t^\alpha(i)$$

Intermediate Goods Sector

Each variety i is produced by a **monopolist** who holds a patent.

Production technology: One unit of capital $\rightarrow$ one unit of intermediate good:

$$X_t(i) = K_t(i)$$

Monopolist's pricing: Each firm faces the downward-sloping demand curve from the final good sector and maximizes profit:

$$\pi_t(i) = P_t(i) X_t(i) - R_t X_t(i)$$

where R_t is the rental rate of capital.

Optimal price (constant markup):

$$P_t(i) = \frac{R_t}{\alpha}$$

The markup is $1/\alpha > 1$ over marginal cost R_t .

Intermediate Goods: Profit and Symmetry

Symmetric equilibrium: All firms face the same demand, so:

$$X_t(i) = X_t, \quad P_t(i) = P_t, \quad \pi_t(i) = \pi_t \quad \text{for all } i$$

Monopolistic profit per variety:

$$\pi_t = (P_t - R_t) X_t = \left(\frac{1}{\alpha} - 1 \right) R_t X_t = \frac{(1 - \alpha)}{\alpha} R_t X_t$$

Using the capital market clearing condition $N_t X_t = K_t$:

$$X_t = \frac{K_t}{N_t}$$

And the expression for R_t :

$$\pi_t = \alpha(1 - \alpha) \frac{Y_t}{N_t}$$

Innovation technology: R&D labour L_R produces new varieties:

$$\dot{N}_t = \theta N_t L_R$$

- $\theta > 0$: R&D productivity parameter
- **Knowledge spillover**: current stock N_t raises R&D productivity
- This “standing on shoulders of giants” effect is central to sustained growth

Value of an invention: The present value of future monopoly profits:

$$v_t = \frac{\pi_t}{r - g_\pi}$$

where g_π is the growth rate of profits (on a BGP, $g_\pi = g_Y = g_N$).

Free entry condition in R&D:

$$\theta N_t v_t = W_t$$

Expected benefit of R&D = wage cost of R&D labour.

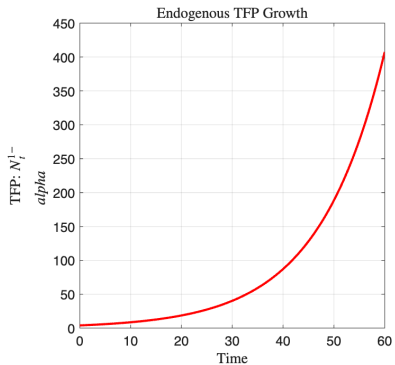
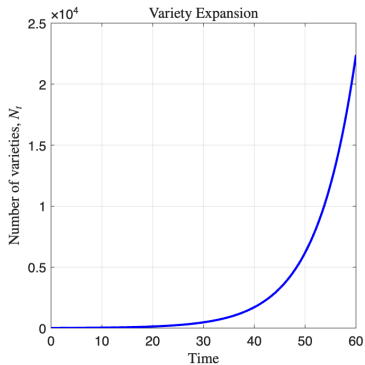
Aggregate production function:

Substituting the symmetric equilibrium ($X_t(i) = K_t/N_t$ for all i) into the production function:

$$Y_t = L_Y^{1-\alpha} \int_0^{N_t} \left(\frac{K_t}{N_t} \right)^\alpha di = L_Y^{1-\alpha} N_t \cdot \left(\frac{K_t}{N_t} \right)^\alpha$$

$$\boxed{Y_t = N_t^{1-\alpha} K_t^\alpha L_Y^{1-\alpha}}$$

- $N_t^{1-\alpha}$ plays the role of TFP: more varieties = higher productivity
- Increasing returns to K and N jointly, but CRS in K and L_Y
- Technology growth $\Leftrightarrow$ growth in N_t



Labour Market Clearing

Total labour supply is allocated between production and R&D:

$$L_Y + L_R = L$$

Balanced growth path:

- L_Y and L_R are *constant*
- Growth rate of varieties: $g_N = \dot{N}/N = \theta L_R$
- Growth rates: $g_Y = g_K = g_C = g_N = \theta L_R$

Interest rate on BGP:

From the Euler equation: $r = \rho + g_C = \rho + \theta L_R$

We need to solve for the equilibrium allocation of labour to pin down L_R .

Solving for Equilibrium R&D

Combine the free entry condition with expressions for v_t , π_t , and W_t :

Step 1: From free entry: $\theta N v = W = (1 - \alpha)Y/L_Y$

Step 2: Value of invention: $v = \pi/(r - g_\pi)$, where $\pi = \alpha(1 - \alpha)Y/N$

Step 3: On BGP: $r - g_\pi = \rho + g_N = \rho + \theta L_R$

Substituting:

$$\theta N \cdot \frac{\alpha(1 - \alpha)Y/N}{\rho + \theta L_R} = \frac{(1 - \alpha)Y}{L_Y}$$

Simplifying:

$$\frac{\alpha \theta}{\rho + \theta L_R} = \frac{1}{L_Y} = \frac{1}{L - L_R}$$

Equilibrium R&D Labour

Solving for L_R :

$$\alpha \theta (L - L_R) = \rho + \theta L_R$$

$$\alpha \theta L - \alpha \theta L_R = \rho + \theta L_R$$

$$\theta L_R (1 + \alpha) = \alpha \theta L - \rho$$

$$L_R = \frac{\alpha}{1 + \alpha} \left(L - \frac{\rho}{\alpha \theta} \right)$$

Equilibrium growth rate:

$$g_N = \theta L_R = \frac{\alpha \theta}{1 + \alpha} \left(L - \frac{\rho}{\alpha \theta} \right)$$

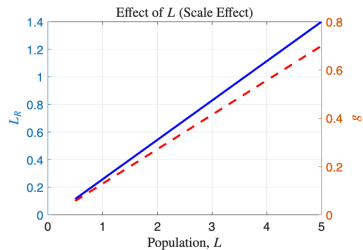
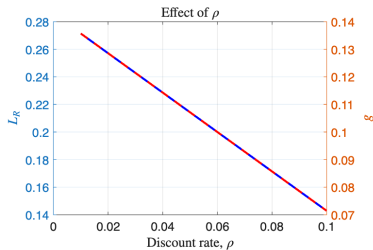
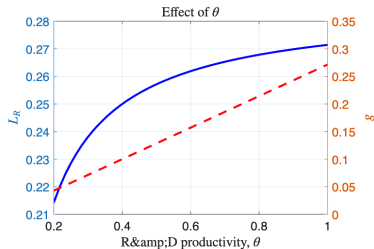
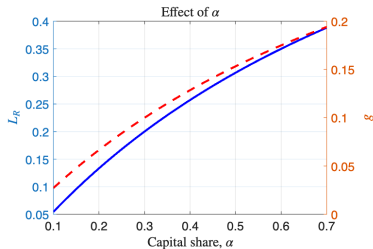
Condition for positive growth: $L > \rho/(\alpha \theta)$.

$$L_R = \frac{\alpha}{1 + \alpha} \left(L - \frac{\rho}{\alpha\theta} \right)$$

Parameter	Effect on L_R	Intuition
$\theta \uparrow$	$L_R \uparrow$	Higher R&D productivity $\Rightarrow$ more R&D
$\alpha \uparrow$	$L_R \uparrow$	Larger monopoly profits $\Rightarrow$ more R&D
$L \uparrow$	$L_R \uparrow$	Larger economy $\Rightarrow$ more R&D (scale effect)
$\rho \uparrow$	$L_R \downarrow$	More impatient $\Rightarrow$ less R&D

Note: The growth rate $g = \theta L_R$ is **increasing in population size L** . This is the **scale effect**: larger countries should grow faster.

Romer Model: Comparative Statics



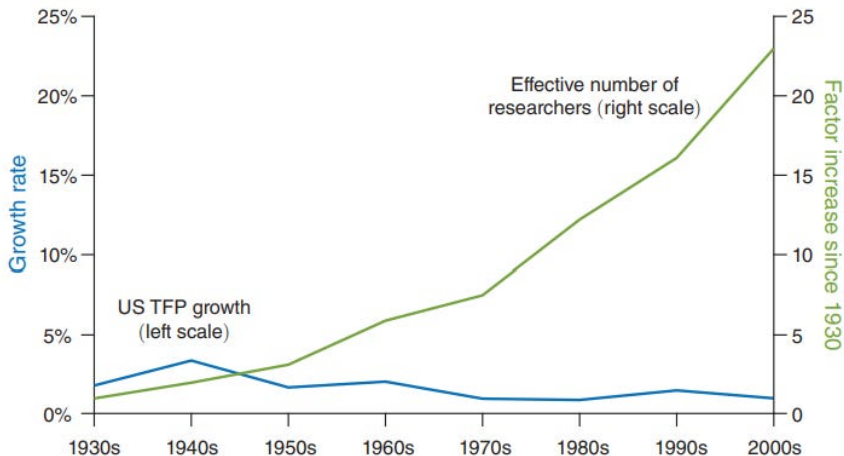
Jones Model

The Scale Effect Problem

Romer model prediction: $g = \theta L_R$, so the growth rate is increasing in population L .

Empirical evidence (Jones, 1995):

- The number of scientists and engineers in the US has grown **dramatically** since 1950
- Yet TFP growth has remained roughly **constant** (or even declined)
- Similar patterns in other advanced economies



The strong scale effect prediction is **rejected by the data**.

Jones (1995): Semi-Endogenous Growth

Modification: Replace the R&D technology $\dot{N} = \theta N L_R$ with:

$$\dot{N}_t = \theta N_t^\phi L_{R,t}, \quad \phi < 1$$

- $\phi < 1$: **diminishing returns** to the existing stock of knowledge in R&D
- $\phi = 1$: Romer's original specification
- $\phi < 1$: “fishing out” effect—easier ideas are discovered first

Interpretation:

- As N grows, it becomes progressively *harder* to discover new varieties
- Each new idea “stands on the shoulders of giants” less effectively
- $1 - \phi$ measures the degree of diminishing returns in R&D

Jones Model: Balanced Growth Path

On a BGP, the growth rate $g_N = \dot{N}/N$ must be constant.

$$g_N = \frac{\dot{N}}{N} = \theta N^{\phi-1} L_R$$

For g_N to be constant with $\phi < 1$:

- $N^{\phi-1}$ is *decreasing* over time (since $\phi < 1$ and N is growing)
- So L_R must be *increasing* to compensate
- This requires population growth: $\dot{L}/L = n > 0$

BGP growth rate:

$$g_N = \frac{n}{1 - \phi}$$

- Growth depends on **population growth rate** n , not population *level* L
- **No scale effect!**
- Growth is “semi-endogenous”: determined by parameters, but $n = 0 \Rightarrow g = 0$

Jones Model: Equilibrium R&D Allocation

Define $S_R \equiv L_R/L_Y$ (ratio of R&D to production labour).

Following the same approach as in the Romer model but with $\phi < 1$ and $\dot{N}/N = g_N = n/(1 - \phi)$:

$$S_R = \frac{\alpha n}{(1 - \phi)\rho + n}$$

Comparative statics:

- S_R is **increasing** in n and α
- S_R is **decreasing** in ρ and in $(1 - \phi)$
- S_R does **not** depend on $L \Rightarrow$ no scale effect

Note: Policy can still affect the *level* of output (and welfare) through S_R , but not the long-run *growth rate*.

Romer vs. Jones: Summary

	Romer (1990)	Jones (1995)
R&D technology	$\dot{N} = \theta N L_R$	$\dot{N} = \theta N^\phi L_R$
Knowledge spillover	Full ($\phi = 1$)	Partial ($\phi < 1$)
BGP growth rate	$g = \theta L_R$	$g = n/(1 - \phi)$
Scale effect	Yes (in L)	No
Population growth	Not needed	Required ($n > 0$)
Policy $\rightarrow$ growth rate	Yes	No (only level)

Trade-off:

- Romer: rich policy implications but counterfactual scale effect
- Jones: consistent with data but long-run growth is beyond policy reach

Social Planner's Problem

Question: Is the market equilibrium level of R&D socially optimal?

A benevolent social planner maximizes household utility subject to:

- Production function: $Y_t = N_t^{1-\alpha} K_t^\alpha L_Y^{1-\alpha}$
- R&D technology: $\dot{N}_t = \theta N_t^\phi L_{R,t}$
- Resource constraint: $L_Y + L_R = L$

Focus on Jones specification ($\phi = 0$ for simplicity):

On the BGP, the socially optimal R&D ratio is:

$$S_R^* = \frac{n}{\rho + n}$$

Social Planner: Setup (Jones, $\phi = 0$)

The **centralised** planner maximises household utility directly subject to the economy's constraints:

$$U = \int_0^{\infty} e^{-\rho t} \ln C_t dt \quad (15.2)$$

$$Y_t = N_t^{1-\alpha} K_t^{\alpha} L_{Y,t}^{1-\alpha} \quad (15.3)$$

$$\dot{K}_t = Y_t - C_t - \delta K_t \quad (15.4)$$

$$\dot{N}_t = \theta L_{R,t} \quad (\phi = 0) \quad (15.5)$$

$$L_{R,t} + L_{Y,t} = L_t = L_0 e^{nt} \quad (15.6)$$

Current-value Hamiltonian (dropping time subscripts):

$$\mathcal{H} = \ln C + \lambda_K (N^{1-\alpha} K^{\alpha} L_Y^{1-\alpha} - C - \delta K) + \lambda_N \theta (L - L_Y) \quad (15.7)$$

where λ_K is the multiplier for $\dot{K}$ and λ_N is the multiplier for $\dot{N}$.

Social Planner: First-Order Conditions

FOCs (Chu, Chapter 15):

$$\frac{\partial \mathcal{H}}{\partial C} = \frac{1}{C} - \lambda_K = 0 \implies \lambda_K C = 1 \quad (15.8)$$

$$\frac{\partial \mathcal{H}}{\partial L_Y} = \lambda_K(1 - \alpha) \frac{Y}{L_Y} - \lambda_N \theta = 0 \quad (15.9)$$

$$\frac{\partial \mathcal{H}}{\partial K} = \lambda_K \left(\alpha \frac{Y}{K} - \delta \right) = \lambda_K \rho - \dot{\lambda}_K \quad (15.10)$$

$$\frac{\partial \mathcal{H}}{\partial N} = \lambda_K(1 - \alpha) \frac{Y}{N} = \lambda_N \rho - \dot{\lambda}_N \quad (15.11)$$

From (15.9), the key relationship between multipliers:

$$\boxed{\frac{(1 - \alpha)\lambda_K Y}{\lambda_N} = \theta L_Y} \quad (15.12)$$

Social Planner: BGP Solution

On the BGP: $g_Y = g_C = g_N + n = 2n$ (since $\phi = 0$, $g_N = n$). (15.13)

Rewrite (15.11) as:

$$\frac{\dot{\lambda}_N}{\lambda_N} = \rho - \frac{(1 - \alpha)\lambda_K Y}{\lambda_N N} \quad (15.14)$$

BGP stationarity:

- From (15.8): $\lambda_K C = 1$ is stationary $\Rightarrow \lambda_K Y$ is stationary
- (15.14) LHS is constant $\Rightarrow \lambda_N N$ must be stationary
- Hence $\dot{\lambda}_N / \lambda_N = -g_N = -n$

Substitute (15.12) into (15.14) with $\dot{\lambda}_N / \lambda_N = -n$:

$$-n = \rho - \frac{\theta L_Y}{N} \implies \boxed{\frac{\theta L_Y}{N} = \rho + n} \quad (15.15)$$

From R&D technology ($\phi = 0$): $g_N = \theta L_R / N = n \Rightarrow N = \theta L_R / n$
(15.16)

Social Planner: Optimal R&D Ratio

Substitute $N = \theta L_R/n$ from (15.16) into (15.15):

$$\frac{\theta L_Y}{\theta L_R/n} = \rho + n \implies n \cdot \frac{L_Y}{L_R} = \rho + n$$

$$\boxed{S_R^* \equiv \frac{L_R^*}{L_Y^*} = \frac{n}{\rho + n}} \quad (15.17)$$

Key insight: λ_N captures the *full* social marginal value of an idea, $(1 - \alpha)Y/N$, not just the monopolist's private return $\alpha(1 - \alpha)Y/N$.

General case ($\phi < 1$): Following the same method with $\dot{N}_t = \theta N_t^\phi L_{R,t}$ and $g_N = n/(1 - \phi)$, the socially optimal ratio becomes:

$$\boxed{S_R^* = \frac{g_N}{\rho + g_N} = \frac{n}{(1 - \phi)\rho + n}}$$

Comparing market equilibrium vs. social optimum (Chu, Chapter 15):

$$S_R = \frac{\alpha n}{\rho + n} = \alpha S_R^*, \quad \text{where } \alpha < 1 \quad (15.18)$$

Since $\alpha < 1$:

$$S_R < S_R^*$$

The market **underinvests** in R&D!

Why? The **surplus appropriability problem**:

- Social marginal product of N : $\frac{\partial Y}{\partial N} = (1 - \alpha) \frac{Y}{N}$
- Private return to innovator: $\pi = \alpha(1 - \alpha) \frac{Y}{N} = \alpha \cdot \frac{\partial Y}{\partial N}$
- Monopolist captures only fraction α of the social benefit
- Positive R&D externality $\Rightarrow$ market failure

Policy intervention: Government subsidizes R&D at rate $\varsigma \in [0, 1]$. The cost of R&D becomes $(1 - \varsigma)W_t L_{R,t}$. The free entry condition becomes:

$$\dot{N}_t v_t = (1 - \varsigma) W_t L_{R,t}$$

Equilibrium R&D ratio with subsidies (Jones specification, $\phi = 0$):

$$S_R = \frac{1}{1 - \varsigma} \cdot \frac{\alpha n}{\rho + n}$$

Optimal subsidy: Set $S_R = S_R^*$:

$$\frac{1}{1 - \varsigma} \cdot \alpha = 1 \quad \Longrightarrow \quad \boxed{\varsigma^* = 1 - \alpha}$$

Given $\alpha \in (1/3, 1/2)$, the optimal R&D subsidy is ~~50~~67%!

Social Optimum in the Romer Model

In the original Romer specification ($\phi = 1$):

Socially optimal R&D labour:

$$L_R^* = L - \frac{\rho}{\theta}$$

Market equilibrium:

$$L_R = \frac{\alpha}{1 + \alpha} \left(L - \frac{\rho}{\alpha\theta} \right)$$

Comparison: $L_R < L_R^*$ always holds, so R&D underinvestment is present in the Romer model as well.

Intuition:

- Intertemporal knowledge spillover: $\dot{N} = \theta N L_R$ means current R&D raises future R&D productivity
- Private entrepreneurs do *not* internalize this social benefit
- This is in addition to the surplus appropriability problem

Schumpeterian Model

Joseph Schumpeter (1942): *Creative Destruction*

- Growth driven by **quality-improving** (“vertical”) innovations
- New innovations **destroy** the market position of incumbents
- “The process of industrial mutation that incessantly revolutionizes the economic structure from within, incessantly destroying the old one, incessantly creating a new one”

Key contributions: Aghion & Howitt (1992), Grossman & Helpman (1991)

Key difference from Romer (1990):

- Romer: growth through **new product varieties** (horizontal innovation)
- Schumpeter: growth through **quality improvement** of existing products (vertical innovation)

Toy Model: Setup (Discrete Time)

Discrete time, myopic agents living for one period. In each period t , the final good Y_t is produced using labour L and a single intermediate good y_t :

$$\boxed{Y_t = (A_t L)^{1-\alpha} y_t^\alpha}, \quad \alpha \in (0, 1)$$

where A_t is the **productivity (quality)** of the intermediate input.

Intermediate good produced by a monopolist using final good 1-for-1: $y_t = X_t$.

Resource constraint:

$$Y_t = C_t + X_t + R_t$$

where R_t is spending on R&D.

Quality ladder: Each period, an entrepreneur can improve the quality of the intermediate good:

$$A_t = \begin{cases} \gamma A_{t-1} & \text{with probability } z_t, \\ A_{t-1} & \text{with probability } 1 - z_t. \end{cases}$$

where $\gamma > 1$ is the **step size** of a quality improvement.

R&D cost function:

$$\boxed{R_t = c(z_t) A_{t-1}}, \quad c(z_t) = \frac{\delta}{2} z_t^2$$

- z_t : innovation intensity (probability of success)
- δ : inversely measures R&D productivity
- Cost proportional to frontier technology A_{t-1} : ideas become harder to find (“fishing out”)

Solving the Model: Equilibrium Production and Profits

Solve by **backward induction**. First, for a given A_t , the final-good producer's inverse demand:

$$p_t = \frac{\partial Y_t}{\partial y_t} = \alpha (A_t L)^{1-\alpha} y_t^{\alpha-1}$$

Monopolist's problem: $\max_{y_t} \{p_t y_t - y_t\}$ (marginal cost = 1).

Equilibrium outcome:

$$y_t = \alpha^{\frac{2}{1-\alpha}} A_t L \quad (\text{quantity})$$

$$p_t = \frac{1}{\alpha} \quad (\text{constant markup})$$

$$\pi(A_t) = \bar{\pi} A_t L, \quad \bar{\pi} \equiv (1 - \alpha) \alpha^{\frac{1+\alpha}{1-\alpha}} \quad (\text{profit})$$

Profit is proportional to effective labour supply $A_t L$, and the markup is $1/\alpha > 1$.

Equilibrium Innovation Intensity

The entrepreneur chooses z_t to maximize expected net benefit:

$$\begin{aligned} \max_{z_t} \{ & z_t \pi(\gamma A_{t-1}) - c(z_t) A_{t-1} \} \\ \implies \max_{z_t} \{ & z_t \gamma \bar{\pi} L - \frac{\delta}{2} z_t^2 \} \end{aligned}$$

First-order condition: $\delta z_t = \gamma \bar{\pi} L$

$$z = \frac{\gamma \bar{\pi} L}{\delta}$$

Innovation intensity increases with:

- Larger market size L (scale effect)
- Larger step size γ (bigger innovation jump)
- Higher markup (lower α)
- More productive R&D (lower δ)

Final output: substituting y_t into the production function,

$$Y_t = \alpha^{\frac{2\alpha}{1-\alpha}} A_t L \quad \implies \quad g_t = \frac{A_t - A_{t-1}}{A_{t-1}}$$

Growth is **stochastic** in a single sector:

$$g_t = \begin{cases} \gamma - 1 & \text{with probability } z, \\ 0 & \text{with probability } 1 - z. \end{cases}$$

Expected growth rate:

$$g = z(\gamma - 1)$$

Long-run average growth = **frequency of innovations** $\times$ **size of innovations**. In a multisector economy, growth becomes deterministic by the law of large numbers.

With Many Sectors

Final good is now produced using a continuum of intermediate inputs, according to the logarithmic production function

$$\ln Y_t = \int_0^1 \ln y_{jt} dj$$

Each intermediate producer

$$y_{it} = A_{it} l_{it}$$

We can go through a similar process. The key point is that

- Some producer innovates and brings their productivity to γA_{jt}
- Some producer j might fail to innovate, the probability is $1 - z$ and productivity remains the same as previously realized.

Aggregate growth will then be deterministic by the law of large number

$$g = z \ln \gamma$$

Schumpeterian Growth: Stochastic Quality Ladder

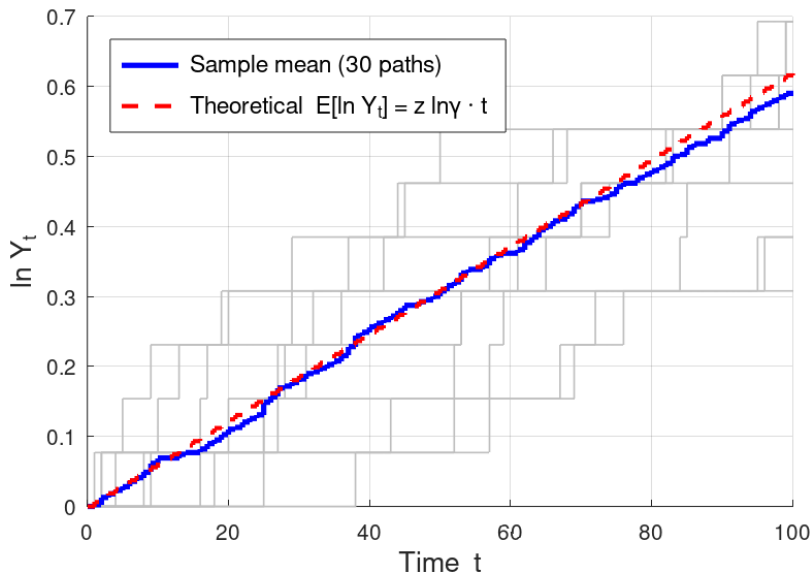
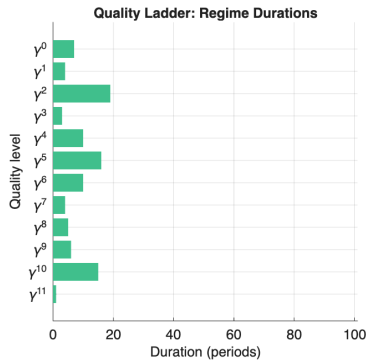
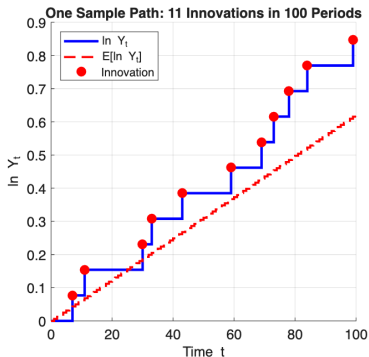


Figure: Simulation



Value of an Invention: Derivation

An inventor who creates a new variety at time t receives the stream of monopoly profits π_s at each instant $s \geq t$. The present discounted value is:

$$v_t = \int_t^{\infty} e^{-r(s-t)} \pi_s ds$$

On the balanced growth path:

- Profits per variety: $\pi_t = \alpha(1 - \alpha) \frac{Y_t}{N_t}$
- Growth rate of profits: $g_{\pi} = g_Y - g_N$
- Since Y and N grow at the same rate on the BGP, $g_{\pi} = 0$
- So $\pi_s = \pi_t$ for all $s \geq t$ (profits per variety are *constant*)

Evaluating the integral:

$$v_t = \pi_t \int_t^{\infty} e^{-r(s-t)} ds = \frac{\pi_t}{r} \quad (\text{since } g_{\pi} = 0)$$

From the Euler equation with log utility, $r = \rho + g_C = \rho + \theta L_R$:

$$v_t = \frac{\pi_t}{\rho + \theta L_R} = \frac{\alpha(1 - \alpha) Y_t / N_t}{\rho + \theta L_R}$$

Intuition: Faster growth $\Rightarrow$ higher r (consumers require more compensation to defer consumption) $\Rightarrow$ *lower* present value of any given profit stream. This is the key link between the household side and the R&D decision.